E-WASTE MANAGEMENT COMPLIANCE ANALYSIS

India's Top Electronics Manufacturers

Based on E-Waste (Management) Rules, 2022

E-WASTE MANAGEMENT COMPLIANCE ANALYSIS

India's Top Electronics Manufacturers

Based on E-Waste (Management) Rules, 2022

1. EXECUTIVE SUMMARY

This document provides a comprehensive analysis of E-Waste Management compliance requirements under India's E-Waste (Management) Rules, 2022 for five leading electronics manufacturers. The analysis covers manufacturers' regulatory obligations, registration requirements, Extended Producer Responsibility (EPR), and environmental compliance standards.

2. SELECTED COMPANIES

The following five major electronics manufacturers have been analyzed:

S.No	Company Name	Key Business Areas
1	Foxconn India	Electronics Manufacturing, Mobile Phones, Computing Devices
2	Tata Electronics	EMS (Electronics Manufacturing Services), Defense, Aerospace
3	Dixon	Consumer Electronics, Smart TVs, Mobile Phones, Appliances
4	Kaynes Technology	EMS, Embedded Systems, Telecom, Computing Devices
5	Centum Electronics	PCB Manufacturing, Aerospace, Defense, Industrial Electronics

3. REGULATORY FRAMEWORK

3.1 Key Regulations

E-Waste (Management) Rules, 2022 – Effective from 01-04-2023 under Environment Protection Act, 1986

Objective: To ensure that E-Waste is managed in a manner which protects health and environment against any adverse effects.

3.2 Scope of E-Waste

E-Waste Definition: Electrical and electronic equipment, including solar photo-voltaic modules or panels or cells, whole or in part discarded as waste, as well as rejects from manufacturing, refurbishment and repair processes.

Coverage: 106 EEEs (Electrical and Electronic Equipment) under seven categories

3.3 E-Waste Composition & Hazards

Valuable Materials: Plastics, Iron, Glass, Aluminum, Copper, Precious Metals (Silver, Gold, Platinum, Palladium), Rare Earth Elements

Hazardous Substances: Lead, Cadmium, Mercury, Hexavalent Chromium, Polychlorinated Bi-phenyls (PCBs), Brominated Flame Retardants (BFR)

4. MANDATORY COMPLIANCE REQUIREMENTS FOR MANUFACTURERS & PRODUCERS

4.1 Extended Producer Responsibility (EPR)

Definition: Responsibility of producers to meet recycling targets through registered recyclers only.

EPR Obligations (Phase-Wise):

Financial Year	Recycling Target (% of E-Waste Generated)
FY 2023-24 & 2024-25	60%
FY 2025-26 & 2026-27	70%
FY 2027-28 onwards	80%

4.2 Registration Requirements

Entities Required to Register on E-Waste EPR Portal:

Entity Type	Definition & Requirements
Manufacturer	Entity with facilities for manufacturing EEEs. Must have valid CTO (Consent to Operate) under Air & Water Act for Green/Orange/Red categories. Must have Authorization under Hazardous and Other Waste (Management) Rules, 2016.
Producer	Any person who manufactures, assembles, or imports EEEs under their own brand. Mandatory registration for all producers selling EEEs listed in Schedule-I.
Recycler	Entity engaged in recycling and reprocessing of e-waste for recovery of precious metals. Must have valid CTO and Hazardous Waste Authorization.
Refurbisher	Any person repairing/assembling used EEEs for extending their working life. Must have CTO and Hazardous Waste Authorization.

4.3 Registration Fee Structure for Manufacturers & Producers

Annual E-Waste Target (MT)	Processing Fee (₹)	Renewal/Amendment
< 50 MT	2,500	₹10,000
50-100 MT	7,500	₹10,000
100-1000 MT	1,50,000	₹10,000
1000-5000 MT	10,00,000	₹10,000
> 5000 MT	15,00,000	₹10,000

4.4 RoHS (Reduction of Hazardous Substances) Compliance

All new EEEs manufactured or sold must comply with maximum concentration values:

- Lead, Mercury, Hexavalent Chromium, PBBs, PBDEs: 0.1% by weight
- Cadmium: 0.01% by weight

Documentation Requirements:

- Self-declaration of RoHS compliance
- Declaration on availability of technical documents (EN 50581/EN IEC 63000:2018)
- Make available for verification by CPCB/SPCBs officials

4.5 EPR Certificate Generation & Fulfillment

Key Metals for EPR Calculation:

- Gold (Au), Copper (Cu), Aluminum (Al), Iron (Fe)

Producer Obligations:

- Fulfill EPR targets only through registered recyclers
- Purchase EPR certificates based on recycled end products
- Maintain records of all purchases and EPR compliance
- Submit quarterly and annual returns to CPCB

4.6 Special Requirements for Solar Photo-Voltaic Modules

- Included in Schedule I coverage
- No recycling target (storage only till 2034-35)
- Manufacturers must store solar waste as per CPCB guidelines

5. SALIENT FEATURES & REQUIREMENTS SUMMARY

5.1 Key Compliance Obligations for All Manufacturers

S.No	Compliance Requirement
1	Register on E-Waste EPR Portal as Manufacturer/Producer within specified timeline
2	Obtain and maintain CTO (Consent to Operate) from SPCB under Air & Water Act
3	Obtain Authorization under Hazardous and Other Waste (Management) Rules, 2016
4	Ensure all new EEEs comply with RoHS limits on hazardous substances
5	Maintain technical documentation on RoHS compliance (EN 50581/EN IEC 63000:2018)
6	Meet phase-wise EPR targets through registered recyclers only (60-80% of e-waste generated)
7	Purchase EPR certificates from registered recyclers to fulfill obligations
8	Maintain detailed records of e-waste collection, dismantling, and recycling
9	Submit quarterly and annual returns to CPCB with recycling data and certifications
10	Pay processing fees and annual maintenance charges (₹5,000 annually for all stakeholders)
11	Undergo third-party audits of compliance and stakeholder records
12	Maintain proper storage and handling facilities for e-waste (especially solar modules till 2034-35)

6. DETAILED COMPLIANCE ANALYSIS: SPECIFIC COMPANIES

6.1 FOXCONN INDIA

Business Profile: Large-scale contract electronics manufacturer; produces mobile phones, computing devices, and consumer electronics.

COMPLIANCE STATUS - RULES THAT MUST BE FOLLOWED:

1. Producer Registration Requirement:

Foxconn, as a manufacturer and seller of multiple EEE categories (mobile phones, computing devices, electronic components), MUST register on the E-Waste EPR Portal as both Manufacturer and Producer.

2. EPR Obligation Compliance:

- Current (FY 2023-24 & 2024-25): Achieve 60% recycling target of total e-waste generated from all products sold
- FY 2025-26 onwards: Scale to 70-80% recycling targets
- For imported used EEEs (if applicable): 100% EPR target

3. RoHS Compliance Declaration:

- Self-declaration that all new products comply with RoHS limits (0.1% for Lead, Mercury, Hexavalent Chromium, PBBs, PBDEs; 0.01% for Cadmium)
- Maintain technical documentation per EN 50581/EN IEC 63000:2018
- Make available for CPCB/SPCB verification

4. Environmental Clearance & Authorization:

- Valid Consent to Operate (CTO) from SPCB for manufacturing facility (if classified as Green, Orange, or Red category industry)
- Authorization under Hazardous and Other Waste (Management) Rules, 2016 from concerned SPCB

5. EPR Certificate Management:

- Purchase EPR certificates only from registered recyclers for all four key metals (Gold, Copper, Aluminum, Iron)
- Maximum certificate purchase: X + 0.05X + Y (where X = EPR obligation, Y = leftover liability)
- Maintain surplus gold certificate limit at ≤ 5%

6. Record Keeping & Reporting:

- Maintain records of all products sold (with quantities and specifications)
- Document all EPR certificate purchases from recyclers
- Submit quarterly returns to CPCB portal with recycling data
- Submit annual comprehensive return with certified data

7. Fee Payment Obligations:

- Initial registration fee: Based on annual e-waste target (likely > 1000 MT: ₹10,00,000 to ₹15,00,000)
- Annual maintenance charge: ₹5,000 per year for all stakeholders
- Amendment/renewal fees: ₹10,000 for any changes in registration

POTENTIAL COMPLIANCE GAPS (Rules that may not be fully followed):

1. Complete Collection Coverage:

Challenge: Foxconn manufactures for multiple brands; tracking product end-of-life across different distribution channels and consumers globally is extremely challenging. EPR rules require producers to track waste generated from their products.

2. Full Recycling Facility Compliance:

Issue: Foxconn may face difficulty ensuring ALL its products are recycled by only registered recyclers. Informal sector e-waste collection remains significant in India.

3. Real-Time Data Verification:

Concern: Portal accepts data from 1st April 2023 onwards. Historical e-waste (pre-2023) may not be fully tracked or validated.

6.2 TATA ELECTRONICS

Business Profile: EMS provider; serves defense, aerospace, industrial, and commercial sectors with complex electronics manufacturing.

COMPLIANCE STATUS - RULES THAT MUST BE FOLLOWED:

1. Manufacturer Registration:

Tata Electronics MUST register as a Manufacturer on the EPR Portal for all EEEs manufactured under Schedule I, including:

- Industrial control equipment
- Telecom equipment
- Computing equipment
- Aerospace/Defense electronics

2. Extended Producer Responsibility for Components:

- As a manufacturer of components and assemblies that go into final products, Tata is responsible for ensuring components comply with regulations
- Must meet phase-wise EPR targets for manufactured products (60-80%)
- Must track e-waste from refurbished or end-of-life defense/aerospace electronics

3. RoHS Compliance for All Products:

- All new components and assemblies must meet RoHS requirements
- Critical for defense/aerospace: Must maintain detailed technical documentation
- Special consideration: Components for equipment placed in market before 01-May-2014 may have exemptions (Schedule IIB), and till 01-April-2028 for certain Schedule IIC items

4. Waste Management Infrastructure:

- Must have valid CTO under Air & Water Act
- Must have Hazardous Waste Authorization
- Applicable for both manufacturing and assembly facilities

5. Recycling Partnership & Collaboration:

- Must partner only with registered recyclers on EPR Portal
- Can collect e-waste across states (no geographic restriction for registered manufacturers/recyclers)
- Must verify recycler credentials before engagement

6. Quarterly/Annual Reporting:

- Submit recycling data for all manufactured products
- Declare key metal recovery (Gold, Copper, Aluminum, Iron)
- Required for both civil and defense electronics manufacturing

7. Fee Obligations:

- Registration fee: ₹15,000 (one-time as Manufacturer)
- Annual maintenance: ₹5,000

POTENTIAL COMPLIANCE CHALLENGES:

1. Complex Supply Chain in Defense/Aerospace:

Issue: Defense equipment may have extended operational life (decades). Tracking end-of-life e-waste from 20-30 year old systems is complex.

2. Multi-Tier Supply Chain Component Tracking:

Difficulty: Tata manufactures components sold to other OEMs who then sell to end-users. Responsibility boundary can be ambiguous under EPR rules.

3. Legacy Equipment Exemptions:

Complexity: Understanding and applying RoHS exemptions for components placed before specified dates requires ongoing compliance tracking.

6.3 DIXON

Business Profile: Consumer electronics brand and manufacturer; products include smart TVs, mobile phones, lighting, appliances, and consumer IoT devices.

COMPLIANCE STATUS - RULES THAT MUST BE FOLLOWED:

1. Producer Registration (Primary Obligation):

Dixon MUST register as Producer on EPR Portal for all consumer electronics sold under Dixon brand, including:

- Smart TVs and display devices
- Mobile phones and communication devices
- Consumer IoT and smart appliances
- Lighting products

2. EPR Obligation for Consumer Products:

- Consumer products generate significant e-waste. Dixon must track and meet recycling targets:
- FY 2023-24 & 2024-25: 60% of generated e-waste
- FY 2025-26 & 2026-27: 70%
- FY 2027-28 onwards: 80%

3. Sales-Based EPR Targets (New Manufacturer/Recent Entrant):

If Dixon is a new entrant/recent sales starter:

- FY 2023-24: 15% of sales in FY 2021-22
- FY 2024-25: 20% of sales in FY 2022-23
- FY 2025-26 onwards: 20% of sales in FY two years prior

4. RoHS Compliance for All Consumer Products:

- All new TVs, phones, appliances, smart devices must meet RoHS limits
- Must maintain compliance documentation (EN 50581/EN IEC 63000:2018)
- Consumer products visibility means higher scrutiny from CPCB/SPCB

5. Consumer-Centric Collection Challenge:

- Must establish mechanisms to collect e-waste from consumers for recycling
- Can tie with retail networks, service centers, or registered recyclers
- Must document collection sources and chain of custody

6. Manufacturer/Producer Dual Registration:

- If Dixon manufactures some products domestically: Also register as Manufacturer
- Requires valid CTO and Hazardous Waste Authorization for manufacturing facilities

7. EPR Certificate Purchase & Compliance:

- Purchase EPR certificates from registered recyclers for all key metals
- GST-linked sales invoices required as proof of recycled metal sales
- Maintain audit trail of all transactions

8. Fee Structure:

- Initial registration: Processing fee based on e-waste target (likely ₹1,50,000-₹10,00,000)
- If new/recent: ₹10,000 initial processing fee
- Annual maintenance: ₹5,000

LIKELY COMPLIANCE GAPS:

1. Consumer Product End-of-Life Tracking:

Major Challenge: Consumer products (TVs, phones) sold across India are often disposed through unorganized channels. Tracking actual e-waste generation and collection is very difficult.

2. Informal Sector E-Waste Dominance:

Issue: Most consumer e-waste in India goes to informal recyclers. Dixon must ensure waste reaches only registered recyclers, which is challenging.

3. Warranty & Return Systems:

Gap: Most consumer products sold don't come back through official warranty channels. Establishing reverse logistics for e-waste is expensive.

6.4 KAYNES TECHNOLOGY

Business Profile: EMS provider specializing in embedded systems, telecom equipment, computing devices, and industrial electronics.

COMPLIANCE STATUS - RULES THAT MUST BE FOLLOWED:

1. Manufacturer Registration:

Kaynes MUST register as Manufacturer on EPR Portal for all EEEs manufactured, including:

- Embedded systems and controllers
- Telecom network equipment
- Computing devices and modules
- Industrial electronic equipment

2. EPR Obligation for Manufactured Products:

- Must meet phase-wise recycling targets for all manufactured products (60-80%)
- Must track e-waste from products sold under own brand or to other OEMs
- Telecom equipment may have longer operational life; must track end-of-life

3. Producer Registration (if also selling under own brand):

If Kaynes sells products directly or through subsidiaries under its own brand, must also register as Producer

4. RoHS Compliance Declaration:

- All new embedded systems and equipment must comply with RoHS limits
- Technical documentation required per EN 50581
- Ready for CPCB/SPCB verification

5. Environmental Certifications:

- Valid CTO from SPCB for manufacturing facility
- Hazardous Waste Authorization from SPCB
- Compliance certificates to be renewed on schedule

6. Recycler Collaboration:

- Partner with registered recyclers for all e-waste management
- Purchase EPR certificates for key metals (Gold, Copper, Aluminum, Iron)
- Maintain documented agreements with recyclers

7. Record Keeping & Returns:

- Maintain comprehensive records of all manufactured products and sales
- Submit quarterly recycling data to EPR Portal
- Annual certified returns with auditor sign-off

8. Fee Obligations:

- Registration as Manufacturer: ₹15,000 (one-time)
- Annual maintenance: ₹5,000

COMPLIANCE CHALLENGES:

1. B2B Supply Chain Complexity:

Challenge: Kaynes sells to multiple OEMs (Telecom companies, industrial equipment makers). Tracking end-of-life of products sold to intermediaries is difficult.

2. Telecom Equipment Lifecycle:

Issue: Telecom network equipment may be in use for 10-15 years. Planning and predicting e-waste generation timing is uncertain.

3. Component-Level Manufacturing:

Ambiguity: When Kaynes manufactures components/modules sold to other companies as parts of their products, responsibility boundaries can be unclear.

6.5 CENTUM ELECTRONICS

Business Profile: PCB (Printed Circuit Board) manufacturer; serves aerospace, defense, industrial, and communication equipment sectors.

COMPLIANCE STATUS - RULES THAT MUST BE FOLLOWED:

1. Manufacturer Registration:

Centum MUST register as Manufacturer for all PCBs and electronic assemblies manufactured for Schedule I EEEs.

2. Special Significance: PCB as E-Waste Component

Critical Importance: The E-Waste Rules specifically note that "printed circuit boards (PCBs) contain the most complex mix of substances" - hazardous and valuable materials are concentrated in PCBs.

- Centum's PCBs, when part of end-of-life equipment, become critical e-waste recycling material
- Must ensure PCBs are designed for recyclability and contain no prohibited hazardous substances

3. RoHS Compliance for PCB Manufacturing:

- PCBs must meet RoHS limits for solder materials, coating, and substrates
- Critical metals (gold, silver, copper in PCBs) must be disclosed for recycling purposes
- Technical documentation on material composition required

4. EPR Obligation for PCB Manufacturing:

- Must meet phase-wise recycling targets for manufactured PCBs (60-80%)
- PCBs sold to aerospace/defense equipment makers: Track end-of-life waste
- Industrial communication PCBs: Monitor through customer channels

5. Environmental & Waste Management Certifications:

- Valid CTO from SPCB (likely Orange or Red category due to chemical processing in PCB manufacturing)
- Hazardous Waste Authorization essential (PCB manufacturing involves hazardous chemicals, solvents, metallic waste)
- Waste water treatment certifications

6. Material Recovery & Recycling Collaboration:

- Partner with registered recyclers who have capability to process PCBs
- Ensure recycled PCBs contribute to key metal recovery (Gold, Copper, Aluminum)
- Purchase EPR certificates based on metal content recovered from recycled PCBs

7. Record Keeping & Documentation:

- Maintain detailed records of all PCBs manufactured (quantities, types, material composition)
- Document material flow from manufacturing to customer to end-of-life
- Submit quarterly and annual returns to EPR Portal

8. Fee Obligations:

- Registration as Manufacturer: ₹15,000 (one-time)
- Annual maintenance: ₹5,000

COMPLIANCE CHALLENGES FOR PCB MANUFACTURERS:

1. Extended Supply Chain Tracking:

Difficulty: PCBs are sold to equipment manufacturers (Aerospace, Defense, Telecom OEMs). Tracking which PCBs end up as e-waste and when is extremely challenging.

2. Defense & Aerospace Equipment Lifecycle:

Issue: Equipment using Centum's PCBs may remain operational for 15-30+ years. Predicting e-waste generation timing is very difficult.

3. Recycler Technical Capability:

Challenge: Not all registered recyclers have specialized facilities to properly process PCBs and recover precious metals. Centum must identify qualified recyclers.

4. Data Confidentiality in Defense Sector:

Constraint: Defense equipment PCBs may not be easily released for civilian recycling due to security/classified material concerns.

7. COMPREHENSIVE COMPLIANCE CHECKLIST - ALL FIVE COMPANIES

Compliance Item	Foxconn	Tata Elec.	Dixon	Kaynes	Centum
Producer Registration	MUST	Conditional	MUST	MUST	No
Manufacturer Registration	If Mfg	MUST	If Mfg	MUST	MUST
EPR Obligation (60-80% target)	REQUIRED	REQUIRED	REQUIRED	REQUIRED	REQUIRED
RoHS Compliance Declaration	REQUIRED	REQUIRED	REQUIRED	REQUIRED	REQUIRED
Valid CTO from SPCB	REQUIRED	REQUIRED	If Mfg	REQUIRED	REQUIRED
Hazardous Waste Authorization	REQUIRED	REQUIRED	If Mfg	REQUIRED	REQUIRED
Quarterly Returns to CPCB	REQUIRED	REQUIRED	REQUIRED	REQUIRED	REQUIRED
Annual Certified Returns	REQUIRED	REQUIRED	REQUIRED	REQUIRED	REQUIRED
Recycler Partnership (Registered Only)	REQUIRED	REQUIRED	REQUIRED	REQUIRED	REQUIRED
Annual Maintenance Fee (₹5,000)	REQUIRED	REQUIRED	REQUIRED	REQUIRED	REQUIRED

8. KEY FINDINGS & RECOMMENDATIONS

8.1 Mandatory Obligations (ALL Companies MUST Comply):

1. Register on E-Waste EPR Portal (Producer and/or Manufacturer)

2. Meet phase-wise EPR recycling targets (60-80% of e-waste generated)

3. Ensure RoHS compliance for all new products

4. Maintain CTO and Hazardous Waste Authorization

5. Partner only with registered recyclers

6. Submit quarterly and annual returns to CPCB

7. Pay registration and annual maintenance fees

8.2 Common Compliance Challenges:

Challenge 1 - E-Waste Collection Tracking: All companies struggle to track actual e-waste generated from products sold, especially in consumer channels.

Challenge 2 - Informal Sector Diversion: E-waste often diverts to informal recyclers outside the registered system, making actual EPR fulfillment difficult.

Challenge 3 - Product Lifecycle Variability: Long-life products (telecom, aerospace, defense equipment) make end-of-life prediction and EPR planning uncertain.

Challenge 4 - Registered Recycler Availability: Limited number of facilities meeting CPCB standards for specialized e-waste (PCBs, hazardous materials).

Challenge 5 - Cross-State Supply Chains: Products sold through multi-state distribution channels complicate collection and tracking.

8.3 Recommendations:

For All Companies:

- Implement robust e-waste tracking systems and periodic audits
- Establish take-back programs and reverse logistics partnerships
- Engage with third-party auditors for compliance verification
- Maintain comprehensive documentation for CPCB/SPCB inspections
- Plan EPR obligations conservatively given market uncertainties

VERIFICATION & RESOURCES:

- E-Waste EPR Portal: https://eprewastecpcb.in
- CPCB Website: http://cpcb.nic.in/e-waste/
- Email (Technical Support): ewaste2.cpcb@gov.in
- Helpdesk: 011-43102467

Document Prepared: September 2026

Based on E-Waste (Management) Rules, 2022 and amendments thereof